

Module 3A – Making Measurements



In our definition of an experiment, we said that experiments use qualitative and quantitative observations.

The distinction we made about the difference between these two kinds of observations is that quantitative observations involve a number or numbers, whereas qualitative observations do not.

A scientific measurement is composed of three parts, two of which are numbers.

Here is an example of a scientific measurement:

$$12.34 \pm .02 \text{ cm}$$

The first part, 12.34, is called the **best guess** or **best estimate**. The second part, $\pm .02$, is called the **tolerance** or **instrument limiting error** (ILE). The third part, cm, is called the **unit**, which in this case is centimeters.

Note: All modern-day scientific measurements are expressed in decimal format. No one uses fractions in scientific measurements.

Most students first encountering a scientific measurement are going to ask:

“Plus or minus, which is it? Make up your mind!”

A reasonable response is to say “follow the instructions.” In this example, following the instructions produces an interval

$$(12.32, 12.36)$$

which is called the **confidence interval**. Understand that this interval is a range of values with the end points (included) being 12.32 and 12.36.

In fact, it is perfectly acceptable to write this scientific measurement as

$$(12.32, 12.36) \text{ cm}$$

The reason why in science we write measurements as a range of values is that we acknowledge that no scientific measurement (really no measurement) can be written with absolute certainty. Nevertheless, by writing this interval, we can assign a probability (like 67% or 100%) that the confidence interval “captures” the true value of the measurement with a confidence (or probability) equal to the percentage assigned to the interval.

For one measurement, we need to find the smallest tolerance (or ILE) that when added to, and subtracted from the best guess produces a confidence interval with an assigned probability of 100%.

If we make many identical measurements, then statistics may allow us to produce a better “best guess” (typically the arithmetic average of all the measurements), and a statistically determined better tolerance (typically something called the standard deviation of the mean, or sigma, σ).

[By many identical measurements, we mean on the order of 100 measurements. Statistics doesn't allow you to do standard deviation calculations for a small “population size” of say 25 measurements.]

A statistically determined best guess and tolerance usually produces a confidence interval with an assigned probability of 67%.

Essentially, all scientific measurements are probability statements. We can't know what the measurement's true value is for sure, but we can say, to the assigned percentage of probability, that it is between the two endpoints (inclusive) of the confidence interval.

In our example above, we are saying that, for the single measurement we made, our best guess as to the measurement made is 12.34 cm (to the level of the precision of our measuring device), but whatever the true value of the measurement is, we are 100% sure that it is between 12.32 cm and 12.36 cm inclusive.

Note that the tolerance used (here for a single measurement) needs to be as small as possible while still producing a confidence interval with an assigned probability of 100%.

Suppose for example that you worked for a traveling carnival, and your specialty was guessing people's heights for say one dollar. If you guessed someone's height as 5 feet, 5 inches, plus or minus 1 foot (or twelve inches), that person would want her dollar back for sure. Your “measurement” of her height in this case would be so crude (or imprecise) as to lack all meaning.

Suppose now that her height is actually 5 feet, 6 inches, and you guessed her height as 5 feet, 5 inches, plus or minus 1 inch. I think she would agree that you “earned” your dollar.

By saying that her height was 5 feet, 5 inches, plus or minus 1 inch, you are saying scientifically that her true height resides within the interval

(64, 66) in, inclusive

with 100% certainty (you only made one guess or one “measurement”). You are saying that her height is 64, 65 or 66 inches, and not 63 inches or 67 inches.

[Carnival employees are quite good at guessing people's heights. Can you figure out how they do it?]

So, for a single measurement, the real question is how to determine what the smallest tolerance (ILE) needed is, so as to produce a confidence interval with 100% probability of capturing the true measurement. For this question, we need to look at the measuring device used to make the measurement.

At this point, it is necessary to distinguish between **analog** and **digital** measuring devices.

Digital devices always have some screen that displays a number as a sequence of digits. The last digit to the right, the one that changes the smallest amount, is called the **least significant digit** (to be explained later).

Here is a picture of a digital measuring device, the digital caliper:

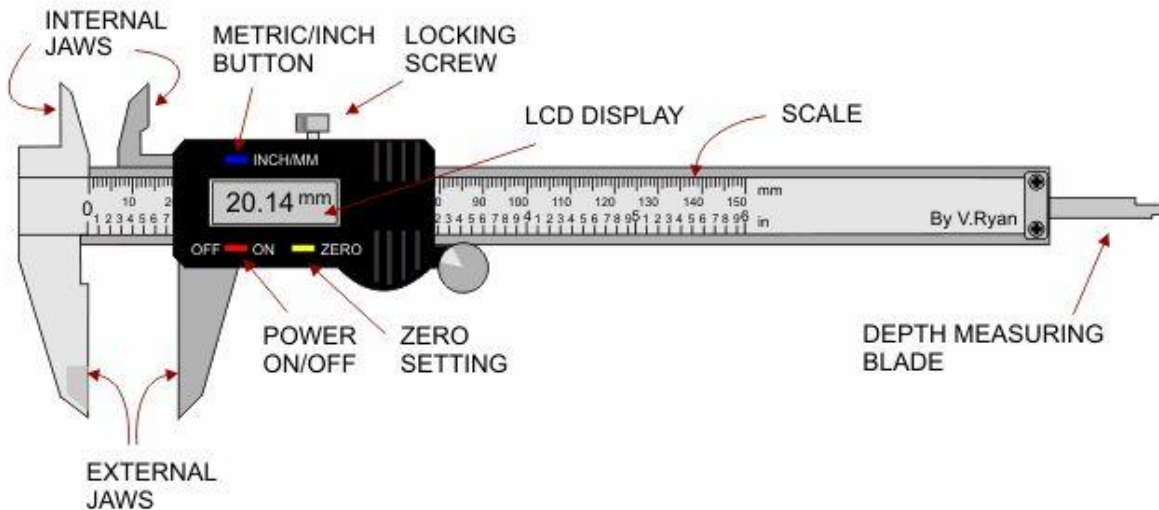


Figure 2

Here we see a measurement of 20.14 millimeters. We say that the measurement has a **precision** to one hundredth of a millimeter (the term precision will be explained later). We also say that the measurement has four **significant figures** (to be defined later), with the least significant digit being the right-most digit, here the number 4.

The digital caliper is just a “glorified ruler” in the sense that it can make length measurements to a higher degree of precision than you personally can just using a regular ruler of the same quality. Put another way, you can’t personally see 100 parts in a millimeter, but the digital caliper can. A millimeter is the thickness of about 10 sheets of paper, so a hundredth of a millimeter would be one tenth the thickness of a sheet of paper!

Today’s digital calipers exhibit readings both as millimeters (mm) and inches (in). For the sake of our discussion here, we will exclusively talk about measurements in mm.

When the jaws of a digital caliper are brought together, the display should say 0.00 mm. If it does not, you should (must) press the “tare” or zero button to set the display to 0.00 mm, otherwise you will be introducing a **zero error** into your measurement (more on this later).

With this caliper, you can make external or internal measurements, using the external and internal jaws respectively. An example of an external measurement would be the length of a metal block. An example of an internal measurement would be the diameter of a cylindrical hole bored into that block.

You can also make a depth measurement, as shown in the illustration above.

Clearly, the digital caliper is a very versatile instrument.

[There is an analog equivalent to the digital caliper, called the [vernier caliper](#), but we won't discuss it here.]

To properly determine the tolerance (or ILE) for the digital caliper, we need to identify its **least count** or **smallest scale division**. For the digital caliper, the least count is .01 mm, because the measurement can only increment up or down by .01 mm (it's [discrete](#)).

The **tolerance or instrument limiting error** for a measuring device is chosen based on the least count of the device. It is [a fraction or percentage of the least count, up to 100%](#).

In the case of the digital caliper, the tolerance or ILE is taken as 100% of the least count. This means that we should quote a scientific measurement using a digital caliper like this:

$$12.34 \pm .01 \text{ mm}$$

Why did we choose the tolerance (or ILE) to be 100% of the least count? Well, two reasons: 1) the repeatability of the digital caliper varies by about .01 mm, and 2) the last digit of a measurement made on a digital caliper often “flickers”, either up one digit or down one digit.

In summary, for the digital caliper:

$$\text{least count} = .01 \text{ mm} \quad \text{tolerance or ILE} = .01 \text{ mm}$$

Note: When stating our measurement, the tolerance (or ILE) and the best guess have the same number of digits to the right of the decimal point. This is to be expected, since to write more or less digits to the right of the decimal point for the tolerance (as opposed to the best guess) would imply that the tolerance has a different level of precision (than the best guess), which doesn't make sense.

Note: Since the tolerance (or ILE) in this case is .01 mm, we are allowed to write the last (or least significant) digit in the best guess as any number between 0 and 9.

A classic [analog](#) measuring device is the **micrometer caliper**, or **micrometer**, or **mic** for short. Analog measuring devices are distinguished by having a ruled surface that you line up an edge, needle, or other marking with.

The mic is a “glorified screw” attached to a ruler. As you turn the [thimble](#) counterclockwise (as viewed from the [ratchet](#) or strain relief clutch side), the [spindle](#) moves to the right (as shown here), away from the (fixed) [anvil](#).

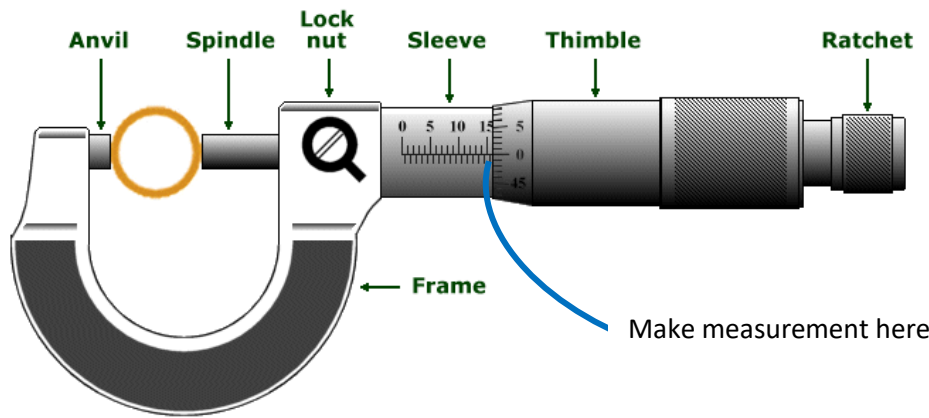


Figure 3

The actual measurement is made on the center line of the ruler visible on the sleeve. One rotation of the thimble goes through the numbers 0 to 50, and the thimble advances 0.50 mm, or 1/2 mm. We see that advance of the thimble because the ruler on the sleeve is marked off in half millimeters, with the half millimeter markings below the regular ruler.

Put another way, if you turn the thimble counterclockwise through two full rotations, the spindle will advance to the right one millimeter.

The least count of the micrometer is clearly .01 mm, the space between adjacent markings on the thimble.

When determining the tolerance (or ILE) of the mic, we usually follow the “one half rule”, that is we take the tolerance to be 1/2 of the least count. So the tolerance (or ILE) of the mic is .005 mm.

The correct way to write the measurement for an analog mic is

$$15.500 \pm .005 \text{ mm}$$

Another possibility would be

$$15.505 \pm .005 \text{ mm}$$

In summary, for the micrometer caliper:

$$\text{least count} = .01 \text{ mm} \quad \text{tolerance or ILE} = .005 \text{ mm}$$

Why do we use the “one half rule” in this case? It’s because conservatively, we say that we can “see” 2 parts in the least count. Equivalently, we say with 100% confidence that the center line of the sleeve is either on a marking on the thimble, or between two adjacent markings on the thimble. By adding and subtracting ½ the least count, we center a least count sized “window” around our best guess.

Note: As with the digital caliper, when stating our measurement, the tolerance (or ILE) and the best guess have the same number of digits to the right of the decimal point.

Note: Since the tolerance (or ILE) for the mic is .005 mm, we are allowed to write the last (or least significant) digit in the best guess only as a 0 or 5.

Note: Some people say they can see 5 parts in the least count of a mic. In that case, the tolerance would be .002 mm, and the possible last digits in the best guess would be 0,2,4, 6 or 8.

Note: When you bring the spindle up to and touching the (fixed) anvil, the zero mark on the thimble should line up with the center line of the sleeve, otherwise you will be introducing a “zero error” in your measurement. Usually, any “zero error” on a mic can be corrected, using a special wrench.

Note: When using a mic, it is critical that you rotate the thimble by turning it with the ratchet (or strain relief clutch). This way, you will not inadvertently “grind” the spindle against one side of whatever it is you are measuring.

The **metric metal ruler** is also an analog measuring device. For rulers that are marked off in mm, the

least count = 1 mm

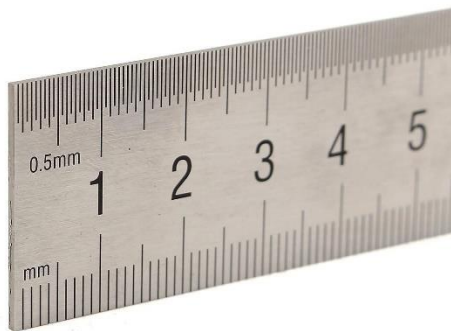
tolerance (or ILE) = 0.5 mm

Thus, scientific measurements with such a ruler would look like:

$13.0 \pm .5$ mm, or

$13.5 \pm .5$ mm, or

$14.0 \pm .5$ mm



Note: For a ruler marked off in mm, the last digit for the best guess and tolerance (or ILE) can only be 0 or 5.

Note: for a ruler with a least count of 1/2 mm, use the 100% rule for the tolerance or ILE. The possible measurements for that ruler would be the same as those given directly above.

Figure 4

Another analog measuring device is the **beam balance scale**.

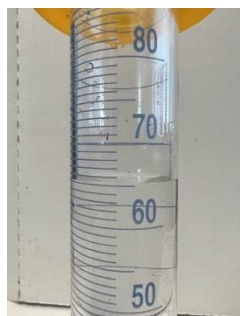
For beam balance scales used in physical science, using the 100% rule, we see that the



Figure 5

least count = 0.1 grams (gm),
and the
tolerance (or ILE) = 0.1 gram

The **graduated cylinder** is also an analog measuring device: it is used to measure liquid volumes.



The graduated cylinder glass side is marked off in **milliliters**. A milliliter (ml) is one cubic cm (1 cm^3), the volume of 1 gram of liquid water at 4 C° .

Normally, since the least count of the graduated cylinder is 1 ml, we would use the 50% rule to get a tolerance for this device.

Therefore, a measurement might be something like

$$13.5 \pm 0.5 \text{ ml}$$

Figure 6

One big problem with using a graduated cylinder is the **meniscus effect**. This is where water creeps up the inside wall of the cylindrical glass tube, making a sort of half “U” shape.



Since most liquid measurements involve changes in liquid level, you can make your liquid level measurement using either the top or the bottom of the meniscus. I prefer the top.

Notice in figure 7 that when we use the top of the meniscus, and viewing the water level straight on (at water level height, to avoid **parallax**, to be discussed below) we can actually see 5 parts in the least count. This means, for the measurement shown, that we can write

$$66.4 \pm 0.2 \text{ ml}$$

Figure 7

I have previously used the expression “least significant digit.” **Significant digits or significant figures** are

all the digits you know for sure, plus one that you “guess” (or is in some way uncertain), when determining your best guess.

So for our initial measurement’s best guess, 12.34, there are four significant digits (or significant figures, or sig figs), with the right-most digit, the 4, being the least significant digit, and the digit that ends up being guessed.

Some important things to know about significant figures:

- Measurements should be expressed in **scientific notation**, i.e.,
a number between 1 and 10, not including 10, with the necessary powers of ten included).
- Measurements stated in **standard notation** (i.e. normal, non-scientific notation), can be confusing as to the number of significant digits in them. For example: leading zeroes in standard notation are not significant, for example:

0.0069 cm has two significant digits,

whereas trailing zeroes in standard notation may be significant, for example if the age of a dinosaur bone is reported to be 75,000,000 years old, how many of the trailing zeroes are actually significant (actually measured)?

- Trailing zeroes in scientific notation are always significant, because otherwise you would not write them. For example, if you wrote that the age of a dinosaur bone in scientific notation is

7.50×10^7 years,

you are stating that the trailing zero was the guessed (and least significant digit).

Significant figures can be confusing at first, and properly using them requires practice and patience.

So far, I have used the expression precision without properly defining it. Actually, there are two terms that go together, precision and accuracy.

Accuracy is

how close you are to the true value of your measurement,

whereas **Precision** is

how many significant figures you can legitimately write in your measurement.

At first blush, it would seem that the better the precision of your measurement, the more accurate it is. For example, if you first write

$$12.34 \pm .01 \text{ cm},$$

and then later, using a more precise measuring device, write

$$12.345 \pm .001 \text{ cm},$$

you would think that the second measurement must be “closer” to the true value of the measurement, and therefore more accurate, because the second measurement has a much smaller confidence interval (or equivalently tolerance).

This however is not necessarily true. You can have a very precise measurement that is inaccurate, if the measuring device used was not properly calibrated. For example, your measuring device has a **zero error** in it that has not been corrected (or at least accounted) for.

The correct understanding of accuracy and precision is stated this way:

It is necessary but not sufficient to be precise in order to be accurate.

A zero error is a so-called trivial systematic error. By contrast, **a non-trivial systematic error** is

an inherent limitation in the equipment or the procedure of an experiment.

A trivial systematic error is only nominally a systematic error. It is really a mistake, a blunder, and should be corrected as a matter of course. Why would you not correct for a zero error?

On the other hand, a non-trivial systematic error is a built-in or inherent limitation in your experiment that you can't really correct for.

For example, if you have to turn a timer on and off by hand (as opposed to electronically), the subsequent time measurement will have systematic anticipation errors built into it (your response to turning on or off the timer may be late or early, if you are responding to a voice command).

Another trivial systematic error is **parallax, or eyeball error**. We will discuss parallax in the First Extended Study (below).

The third part of a scientific measurement, the unit, deserves some attention as well. **A physical unit** is

a standardized amount of a physical quantity.

You need units in commerce so that people will agree how much of something was involved in a transaction (like buying meat or vegetables in a grocery store). The same is true in science experiments.

There are two systems of units in use in science today:

- the **US-English System** (sometimes erroneously referred to as the Imperial System), and
- the **Metric System** (which comes in two varieties, to be discussed shortly).

In some science books, the US-English system is not even discussed, because it is only really used in the United States (there is some limited residual use of a version of this system, called the **Imperial System**, in the United Kingdom. See Homework problem 1.)

The Metric system of units (also called the SI or [Système International](#) system) comes in two varieties, depending on the kind of science being performed. There is the MKS system (which stands for meters, kilograms, seconds), and the CGS system (which stands for centimeters, grams, seconds). In this course we will focus almost exclusively on the MKS metric system. In chemistry, the CGS system is predominantly used.

The most frequently seen physical quantities in science measurements that use units are:

[length, time, mass, energy, force, and charge](#)

Here is a table of those quantities' units for both the US-English and MKS metric systems:

Quantity	US-English	MKS
Length	Foot (ft)	Meter (m)
Time	Second (sec)	Second (sec)
Mass		Kilogram (kg)
Energy	Foot-pound (ft-lb)	Joule (J)
Force	Pound (lb)	Newton (N)
Charge	Coulomb (C)	Coulomb (C)

Table 1

Note: There is a reason why no unit of mass is given here for the US-English system. This will be explained in the homework below. There is disagreement about this issue.

In order to use units either within a system or between the two systems, it is necessary to have **conversion factors**. These are

[ratios equal to the number one, that are used to change one particular unit to another.](#)

Examples of conversion factors within the US-English system of units are:

$$12 \text{ inches} / 1 \text{ foot} = 1$$

$$3 \text{ feet} / 1 \text{ yard} = 1$$

$$5,280 \text{ feet} / 1 \text{ mile} = 1$$

Examples of conversion factors within the MKS metric system of units are:

$$1000 \text{ millimeters} / 1 \text{ meter} = 1$$

$$1000 \text{ meters} / 1 \text{ kilometer} = 1$$

$$1000 \text{ grams} / 1 \text{ kilogram} = 1$$

Examples of conversion factors linking the US-English and the MKS metric systems of units are:

$$2.54 \text{ centimeters} / 1 \text{ inch} = 1$$

$$39.37 \text{ inches} / 1 \text{ meter} = 1$$

$$1.609 \text{ kilometers} / 1 \text{ mile} = 1$$

$$2.205 \text{ lbs} / 1 \text{ kilogram} = 1$$

Note: This last conversion factor is not considered “kosher” because it changes a unit of force in the US-English system to a unit of mass in the MKS metric system. Nevertheless, it is very convenient to use, so we will employ it.

Conversion factors for time are shared between the two systems of units:

$$60 \text{ seconds} / 1 \text{ minute} = 1$$

$$60 \text{ minutes} / 1 \text{ hour} = 1$$

$$24 \text{ hours} / 1 \text{ day} = 1$$

$$365.25 \text{ days} / 1 \text{ year} = 1 \text{ ([Julian Calendar](#) approximation)}$$

First Extended Study: Another kind of trivial systematic error is **parallax** or **eyeball error**. This error only occurs when using your eyes with analog measuring devices.

Parallax is a visual phenomenon whereby nearby objects appear to shift position relative to more distant objects, depending on which eye is being used for viewing.



Figure 8

In the picture to the left, the fir tree appears to shift its position relative to the background (more distant) mountains.

You can try parallax out by simply placing your thumb perpendicularly out from your nose, and looking at something on the wall. Close one eye and make a note of what you see.

Now open that eye and close the other: everything shifts to one side or the other. As you alternate opening and closing one eye at a time, you see the background objects shift

from side to side relative to your thumb.

When using an analog measuring device where you are comparing an edge, or needle (or similar) with a ruled surface, your measurement can easily “shift” erroneously to one side, depending on how you line your eyes up with your measurement, how good your eyes are, and whether one eye has better vision than the other.

If you have equally good vision in both eyes, make sure your nose lines up with the actual measurement, and open and close your eyes sequentially. The measurement should move the same amount from side to side.

If you have “one good eye”, use just that eye, and make sure that eye is directly over your measurement.

Second Extended Study: When performing an experiment, it is quite common to have to multiply two measured quantities together, or divide one measured quantity by another measured quantity.

For example, you might measure a thin rod’s mass using a balance beam scale, and then measure the thin rod’s length, using a digital caliper.

Suppose your results are:

Mass = $m = 12.3 \text{ gm}$

Length = $L = 29.64 \text{ mm} = 2.964 \text{ cm}$

Say you want to calculate the thin rod's mass per unit length, or **linear mass density** (D), given by the formula

$$D = m / L \quad (2)$$

On your calculator, this mathematical operation produces a linear mass density of

$$D = 4.149797571 \text{ gm/cm},$$

a number composed of 10 digits.

But your mass measurement had only three sig figs and your length measurement had four sig figs. Surely a calculator cannot produce extra significant digits out of thin air.

So how many digits are we allowed to write for the ratio D?

The answer is three, which is the lesser of three and four. This is because the uncertainty of the least significant digits of your two factors (m and L) affect (or taint) all but two of the digits that result from the calculation (only the left-most 4 and 1 are "untainted" by the two factors' least significant digits).

Since your answer can only have digits that are known for sure plus one that is guessed, you can write just three digits (i.e., include one tainted digit), so your final answer has to be

$$D = 4.15 \text{ gm/cm}$$

HOMEWORK

1. Imperial Units

I mentioned that the US-English system of units is sometimes erroneously called the Imperial system of units.

The Imperial system of units is an antiquated version the English system of units that we still use in the United States (with significant modifications, hence the expression US-English).

For the most part, the UK today uses the Metric system, but with a few exceptions:

- Distances and speeds are measured in miles and miles per hour respectively;
- Almost all liquids [including gasoline] are sold in liters now, but you still buy pints [20 fl oz, Imperial] or half-pints of beer and certain other beverages;
- Power is generally measured in kW, though occasionally folks still use BTUs.

An interesting aside:

The British only decimalized their money in 1971, so that 100 pennies make up 1 pound. Prior to that time, 12 pennies made up a shilling, and 20 shillings made up a pound (the system being known as pounds, shilling and pence - 240 old pennies to the old pound).



Figure 9

At the time of "decimalization" as it was known, some of the old coins could be used to represent the new ones, until they left circulation. For example, the old 1 shilling coin, worth 12 old pennies became equivalent to a 5 penny piece.

As a result of this monetary conversion, the expression "what's that in old money?" entered the British vernacular to describe a situation where someone was trying to correlate a new piece of information to

something they already knew.

Anyways, Imperial units are at least still [alive and well](#) in British pubs. An Imperial pint is 1/8 of an Imperial gallon, with an Imperial gallon consisting of 160 Imperial ounces (hence one Imperial pint equals 20 Imperial ounces).

By contrast, a US pint is 1/8 of a US gallon, with a US gallon consisting of 128 Yankee ounces.

To make sense of all this, we need to use liters as a conversion factor. A liter is 1000 milliliters, or 1000 cubic centimeters of liquid measure. One gram of pure water occupies one cubic centimeter or one milliliter at exactly 4 degrees Celsius, regardless of external pressure.

It turns out that one Imperial ounce is 28 ml, whereas one US ounce is 30 ml. Thus one Imperial pint is 560 ml, whereas one US pint is only 480 ml.

Presumably, this is why British soccer fans get so drunk at their football matches: they are drinking 17% more beer per pint than their US counterparts.

2. The US-English Unit of Mass



Figure 10

In our analysis of the two systems of units, we left the slot in the units table for the US-English unit of mass conspicuously blank.

It turns out that there are two subsystems within the US-English system of units for mass and force.

Remember from Module 1 that

Weight = mass x acceleration due to gravity, or symbolically,

$$W = m \times g \quad (1)$$

The constant g in the MKS metric system (known as the acceleration due to gravity) happens to be roughly 9.8 m/sec^2 here on Earth. Converting this number to US-English units, g is (roughly) 32 ft/sec^2 .

So in the US-English system of units, if we use pounds ([lbs](#)) as a unit of mass, we must use a different unit for force. That unit would be [poundal](#).

Using the formula above, we say that

one pound mass weighs 32 poundals.

As far as we can tell, no one uses poundals anymore except old-time rocket engineers.

Now if we use pounds in the US-English system of units as a unit of force, then we must use a different unit for mass. That unit is [slug](#).

Using the formula above, we say that

one slug mass weighs 32 pounds.

You may find this unnecessarily confusing.

In this course we will exclusively use pounds force, and never again mention slugs mass. Instead we will use the “non-kosher” conversion factor

$$2.205 \text{ lbs} / 1 \text{ kilogram} = 1$$

to compare mass in the MKS metric to weight in the US-English systems of units.

For example, a 50 kg person has a metric weight of

$$W = m \times g = 50 (9.8) = 490 \text{ Newtons (N)}$$

Using our “non-kosher” conversion factor, we see that a 50 kg person weighs

$$50 \text{ kg} (2.205 \text{ lb} / 1 \text{ kg}) = 110.25 \text{ lbs}$$

Note: Our “non-kosher” conversion factor can also be written

$$454 \text{ gm} / 1 \text{ lb} = 0.454 \text{ kg} / 1 \text{ lb} = 1.$$

This is precisely what is written on every one pound box of cookies in your house.

3. Propagation of Error

This analysis is **only for advanced students** with a higher level of math skills.

In the second extended study above, we calculated the **best guess** for a thin rod’s **linear mass density**, based on a measured mass and a measured length. To review:

$$\text{Mass} = m = 12.3 \text{ gm} \quad \text{Length} = L = 2.964 \text{ cm} \quad D = 4.15 \text{ gm/cm}$$

D was rounded to three significant figures, to reflect that the two factors in the ratio calculated were measured quantities, and a mere calculation cannot add precision.

But D should be stated scientifically, meaning with a tolerance or ILE. How do we calculate (i.e. propagate) that tolerance or ILE?

To review and to amplify our notation, the scientific way to write the single mass measurement is

$$m = 12.3 \pm 0.1 \text{ gm} = \bar{m} \pm \Delta m$$

and the scientific way to write the length measurement is

$$L = 2.964 \pm .001 \text{ cm} = \bar{L} \pm \Delta L$$

The raw (unrounded) value for the linear mass density is

$$D = 4.149797571 \text{ gm/cm}$$

To get the “propagated” value for ΔD , we need to use this formula (underived):

$$\Delta D = D [(\Delta L / \bar{L}) + (\Delta m / \bar{m})]$$

The result is:

$$\begin{aligned}
\Delta D &= 4.149797571 [(.001 / 2.964) + (0.1 / 12.3)] \\
&= 4.149797571 [.000337382 + .008130081] \\
&= 4.149797571 [.008467463] \\
&= .035138257 = 0.04
\end{aligned}$$

rounded up to two digits to the right of the decimal point.

We write our final answer as

$$D = 4.15 \pm .04 \text{ gm/cm}$$

Note: There is another way to get ΔD . You basically calculate the largest possible value for D, the smallest possible value for D, take the difference, divide by 2, and round up to two digits to the right of the decimal point.

For this problem,

$$D \text{ minimum} = 12.2 / 2.965 = 4.114671164, \text{ and}$$

$$D \text{ maximum} = 12.4 / 2.963 = 4.184947688$$

The difference between D maximum and D minimum is 0.070276524,
which divided by 2 equals

$$\Delta D = .035138262 = .04 \text{ (rounding up),}$$

which is almost exactly the same answer.

4. Using Conversion Factors

Use the conversion factors given above to

- change 60 miles per hour (mi/hr, or mph) to feet per second (ft/sec, or fps)
- change 100 kilometers per hour (km/hr, or kph) to miles per hour
- change 60 kilometers per hour to meters per second (m/sec)
- determine many seconds there are in an hour
- Determine how many seconds are in one synodic day, the Earth revolves one time, relative to the “fixed” stars (23 hours, 56 minutes and 4 second seconds).

ASTRONOMY SUPPLEMENT

Measurements made in astronomy are fundamentally different than measurements made here on Earth for science or industry, the reason being that

astronomical measurements are very large, typically with significant uncertainties associated with them.

In astronomy, we use these **units of distance**, ranked from smallest to largest:

astronomical unit (au)

light-year (ly)

parsec (pc)

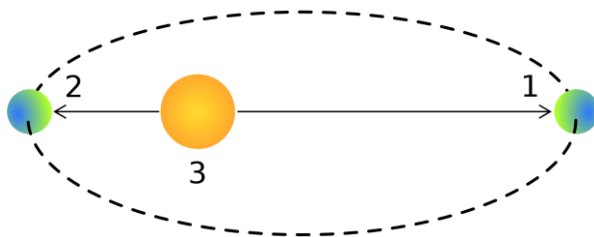


Figure 11

The **astronomical unit** is

the average distance between the Sun and the Earth.

The orbit of the Earth around the Sun is not completely circular. Instead, it is an [ellipse](#), with eccentricity of 0.0167.

When the Earth is closest to the Sun (at [perihelion](#), position 2), the Earth-Sun distance is 91,403,554 miles. This occurs during Winter in the Northern Hemisphere.

When the Earth is farthest away from the Sun (at [aphelion](#), position 1), the Earth-Sun distance is 94,403,554 miles. This occurs during Summer in the Northern Hemisphere.

The value of the astronomical unit comes from using [Kepler's Third Law](#), which connects the periods of each planet (their individual years) to the average distance the planet is away from the Sun, expressed in multiples of the astronomical unit.

Prior to the 20th century, the astronomical unit (au) was poorly known. In 1716, the astronomer [Edmond Halley](#) proposed using the [transit of Venus](#) to determine the au. The 1761 transit of Venus (after Halley's death) subsequently estimated the au to be 95 million miles, a respectable result, compared to our current estimate of 93 million miles (obtained by bouncing radar off Venus and other techniques).

The **light-year** is

the distance light travels in the vacuum of space over the period of one year.

Since light travels 186,000 miles per second, a light-year is a considerable distance. Since light takes 500 seconds to travel from the Sun to the Earth, we say that one

au is equal to 500 light-seconds.

Using conversion factors, it is easy to show that

the light-year is equal to 63,241 au, and

the light-year is equal to 5.879 trillion miles.



Figure 12

In case you are thinking that a light-year is a relatively small distance, it is easy to show that with the currently most advanced spacecraft flying techniques (used for the Parker Solar Probe), it would take a phenomenal **6,619 years**, travelling at **430,000 miles per hour**, to travel to the nearest star (other than our Sun), [Proxima Centauri](#), located a mere **4.24 light-years** from Earth.

Buckle up for a long ride!

Many professional astronomers prefer using the astronomical distance unit known as the parsec. A celestial body is one **parsec** away from Earth

if it subtends two arcseconds angle of parallax shift, when viewed from opposite sides of Earth's orbit.

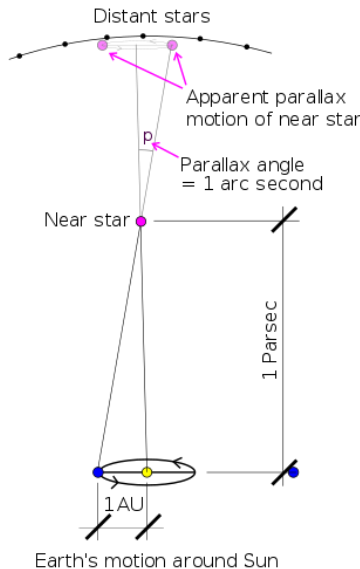


Figure 13

Another way of putting this is that the average radius of Earth's orbit will subtend one arcsecond of parallax shift of position of the star in question, relative to background stars much farther away, if that star is one parsec away.

One parsec (pc) is equal to 3.26 light-years.

So the distance to Proxima Centauri is almost exactly 1.3 parsec.

Sometimes people confuse the distance unit parsec with a unit of time. This was [apparently the case](#) in the "Star Wars: A New Hope" movie.

In astronomy, we use a **hierarchy of measurements techniques** (called the [cosmic distance ladder](#)), with the shorter distance measuring techniques logically being more precise than the longer distance measuring techniques.

From most precise (shortest distances) to least precise (longest distances), they are:

parallax (previously discussed, but reviewed here)

variable stars, used with the inverse square law

type 1a supernovae, used with the inverse square law.

Note: There are a number of lesser known astronomical distance measuring techniques, but we will not discuss them here (see the link directly above).

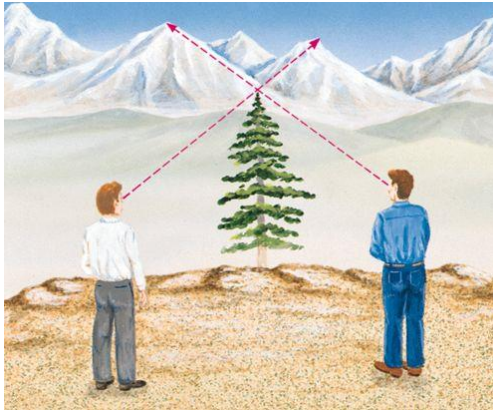


Figure 8

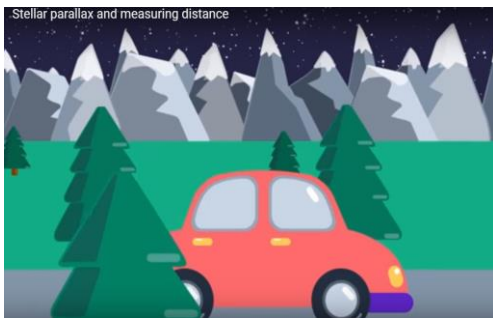
Parallax is

a visual phenomenon whereby nearby objects appear to shift position relative to more distant objects.

In the picture to the left, the fir tree appears to shift its position relative to the background (more distant) mountains, depending on whether the left or right person is viewing the tree.

You can try parallax out by simply placing your hand perpendicularly out from your nose, and

looking at something on the wall. Close one eye and make a note of what you see. Now open that eye and close the other: everything shifts to one side or the other, as you alternate opening and closing one eye at a time.



Video 1

The phenomenon of parallax is used in astronomy as a distance measuring technique. Called **stellar parallax**, or trigonometric parallax, it is quite accurate, but limited to "close in" stars. Currently, we use the [Hubble Space Telescope](#) and stellar parallax to measure distances to stars as far out as **10,000 light-years**. [Click](#) the image to the left to see a video on how stellar parallax works.

Cepheid variable stars are stars that [vary in brightness](#) (or luminosity) in a regular or periodic fashion. Since the maximum brightness and the period of a cepheid star are linearly related, it is possible to determine how far away the star is.



Figure 14

Once we measure the period of a cepheid star (brightest to dimmest to brightest), we can calculate the star's intrinsic brightness. If we then measure the star's apparent brightness, we can use the inverse square law to determine how far away the star is.

This relationship between a cepheid star's period and intrinsic brightness was first discovered in 1912 by "Harvard calculator" [Henrietta Leavitt](#) (image above) and promptly calibrated one year later by astronomer [Einar Hertzsprung](#).

This allowed astronomer [Edwin Hubble](#) to then calculate the distance to cepheid variable stars in [M31](#), then known as the Andromeda Nebula, proving that M31 was not a local "gas cloud," but a far more distant "star city," or galaxy, on a par with our own galaxy, the [Milky Way](#).

Cepheid variable stars constitute the first (and possibly best known) of astronomy's **standard candles**, which are

[astronomical distance measuring techniques that utilize light to determine distances relatively precisely.](#)

Cepheid stars come in two varieties, and have their own calibration constants. The periods of cepheid stars can vary from a few days to two months.

Cepheid stars allow astronomers to measure out to distances of **30 to 40 million parsecs** (Mpc), a good 100 times farther out than the best stellar parallax measurements.

The best known cepheid star is the [North Star](#), Polaris Aa.

Note: The **Inverse Square Law** is a [standard law in optics](#) that states that [the intensity of an omnidirectional light source falls off with the distance squared.](#)

[The Inverse Square Law will be discussed in Module 6A.]

A **supernova** is an [exploding star](#). Supernovae are more than one supernova.



A star explodes when its "corpse" mass -- the mass it has when it runs out of elements to fuse and simply collapses under its own weight -- exceeds 1.44 times the current mass of our star, the Sun. This limit is called the [Chandrasekar Limit](#).

Our Sun -- when it dies about 5 billion years from now -- will not explode, but will instead collapse into a very dense, bright and hot object called a [white dwarf](#).

Figure 15

When a white dwarf is part of a binary (or two) star system, it can, if close enough, “steal” mass from its companion star. The white dwarf star’s mass slowly increases, until its mass is 1.44 solar masses, the Chandrasekar Limit.

The white dwarf star then blows up in a distinctive supernova called a type1a supernova. The type 1a supernova

gives off a characteristic “light curve” whose maximum brightness is precisely known.

We can then use the inverse square law to determine how far away the type 1a supernova was when it happened.

In exploding, the supernova creates elements more massive than iron, and collapses into a new type of “dead” star, a neutron star.

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